Ari Patel

Computer Engineering Student at Arizona State University - Embedded Systems, FPGA Design, and Edge AI

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SUMMARY

Computer Engineering student with hands-on experience in embedded systems, FPGA design, and robotics applications. Demonstrated expertise in developing firmware for microcontrollers (STM32, ESP32), implementing digital logic designs on FPGAs, and integrating control systems. Completed projects spanning motor control, sensor networks, computer vision, and secure embedded systems. Strong foundation in both hardware and software aspects of embedded product development.

EXPERIENCE

Manufacturing Engineering Intern
Acme Robotics

PROJECTS

FPGA Traffic Controller and Pedestrian Safety Core 2025-09 – 2025-11
A Verilog finite-state-machine traffic controller with synchronized inputs, safe fallback states, and timing closure notes.
Technologies: Verilog, FPGA, Digital Logic, FSM, Testbenches, Timing Analysis

RV32I Pipeline Simulator and Hazard Visualizer 2025-10 – 2025-12
A five-stage RISC-V simulator that visualizes forwarding, stalls, branch flushes, and cycle-level benchmark behavior.
Technologies: RISC-V, C++, Python, Computer Architecture, CLI Tools, CSV Analysis

Sun Devil Rover Telemetry Stack 2026-01 – 2026-03
A rover telemetry pipeline combining CAN packets, Raspberry Pi logging, and a real-time dashboard for field debugging.
Technologies: Python, ROS 2, CAN Bus, WebSockets, Raspberry Pi, Telemetry

ESP32 Smart Dorm Sensor Mesh 2025-11 – 2026-02
A low-power ESP32 mesh that reports room comfort, battery state, and calibration health through MQTT.
Technologies: ESP32, FreeRTOS, MQTT, C, Python, Grafana, Low Power

BLDC Motor Driver and PID Test Bench 2026-02 – 2026-04
A motor-control bench with STM32 firmware, current sensing, PID tuning, and oscilloscope-driven bring-up notes.
Technologies: STM32, C, PID Control, PWM, ADC, KiCad, Oscilloscope

FPGA UART Logic Analyzer 2026-03 – 2026-04
A small FPGA capture core that decodes UART traffic into a FIFO and verifies output with a Python golden decoder.
Technologies: Verilog, FPGA, UART, Python, FIFO, Signal Capture

Edge Vision Occupancy Counter 2026-01 – 2026-03
A Raspberry Pi computer-vision prototype that counts room entries locally and publishes privacy-friendly aggregate events.
Technologies: Python, OpenCV, Raspberry Pi, Edge AI, SQLite, Metrics

Secure OTA Bootloader for Cortex-M 2026-04 – 2026-05
A firmware update flow with signed images, anti-rollback checks, A/B slots, and failure recovery notes.
Technologies: C, Cortex-M, Firmware Security, Ed25519, Flash Memory, Threat Modeling

EDUCATION

BSc Mechanical Engineering
State University

SKILLS

Verilog, SystemVerilog, FPGA, RISC-V, C, C++, Python, Embedded Linux, FreeRTOS, ESP32, STM32, PCB Design, KiCad, ROS 2, OpenCV, MQTT, CAN Bus, Control Systems, Signal Processing, GitHub Actions